

OpenFOAM Extensions

Walter Dal'Maz Silva

Contents

1	OpenFOAM Extensions	5
1.1	Compilation methods	5
1.2	Running tests	5
2	Contributing	7
2.1	Directives	7
2.2	Wishlist	7
2.3	Building the docs	7
3	Changelog	9
3.1	2026-09-16	9
I	Applications	
4	jmakLaplacianFoam	13
II	Libraries	
5	Lagrangian	17
5.1	Parcels	17
5.1.1	Extended Thermo Parcel	17
6	Thermophysical Models	19
6.1	Thermophysical Properties	19
6.1.1	Polynomial Solid Properties	19
6.1.2	Tabulated Solid Properties	20
III	Samples	
7	JMAK Solid Kinetics	25
8	Silica Particle Melting	27

1. OpenFOAM Extensions

Curated OpenFOAM 13 directory with additional workflow tools.

1.1 Compilation methods

It is possible to build the whole extensions library or isolated features, depending on your needs. The following logic is applied throughout this sources directory:

- If a directory provides an `Allwmake` file, it allows to compile the whole tree below it; this is the case for the root directory, for instance.
- If a directory of `src/` provides a subdirectory called `Make/`, then it supports the individual library build which is done by running `wmake libso` from that directory.

All builds are written to `$FOAM_USER_LIBBIN`.

1.2 Running tests

Some libraries may provide test programs under a `test/` sub-directory, and all of them are structured in the same way. The following instructions indicate how to build, run, and clean those directories.

To build and run the test suite:

```
cd test
./Allrun
```

To clean test build artifacts:

```
cd test
./Allclean
```


2. Contributing

2.1 Directives

- Comments added by this repository to explain changes performed to base libraries are tagged with `// WWW:` so that they can be tracked in the future.
- Whenever possible/applicable, copy the original `files` list from the library you are modifying to use as a starting point. Remove or comment out files that are not implemented in this repository. Modify the base path of `LIB` to use `LIB = $(FOAM_USER_LIBBIN)/<your-library-name>` so that the path is writable.
- The same apply to `options`, which requires a few extra guidelines concerning its declared variables:
 - `EXE_INC`: keep the original list untouched, add the local include directories in the end. Sometimes the library will fail to compile because the original list was lacking something (in a library as OpenFOAM it probably worked because the missing files were sourced elsewhere during compilation and the authors never had to add them), append these in the end so that we can distinguish them from the ones already in the list.
 - `LIB_LIBS`: if the project compiles but on run-time there is a missing linking library, but proceed as for `EXE_INC` by listing the library in the end. It might be tricky to identify where the functionality comes from, maybe start by `ls $FOAM_LIBBIN` and look for possibly related files.

2.2 Wishlist

- ☐ Unification/generalization of sub-classes of `solidProperties` as a new `extendedSolidProperties`; this can then be used as a replacement to `solidProperties` in the `extendedThermoParcel` framework (possible need to generalize the cloud definition).

2.3 Building the docs

A `Makefile` is provided in the root directory: running `make` will generate a local documentation, `make publish` can be used by admins to publish to GitHub Pages, and `make clean` strips the generated outputs and intermediate files from the folder.

3. Changelog

3.1 2026-09-16

- Adapted `extendedThermoCloud` to upcast the solid object to a `tabulatedSolid` as a prototype for temperature dependent properties. Currently this supports a single species and need to be generalized in the future.
- Unified the build of all libraries under a single `Allwmake` at the sources root, following the full library standard. Added compilation instructions at multiple levels to `README.md`
- Added `extendedThermoCloud` support by extending `thermoCloud`; this extension if fact only needed working with `thermoParcel` as the physics was found to be at the parcel level. Added related library sources under `lagrangian/parcel/{cloud,parcels}`.

I

Applications

4	<code>jmakLaplacianFoam</code>	13
---	--	----

4. jmakLaplacianFoam

Solver for time-temperature reactive heat release in solids. Implements differential Johnson–Mehl–Avrami–Kolmogorov (JMAK) reaction progress kinetics for non-isothermal process as proposed by [Mittemeijer et al.](#).

In the future, it would be desirable to have the following features implemented:

- ☐ Handle non-linearities in thermal properties.
- ☐ Generalization for an arbitrary number of reactions.
- ☐ Implement `solidThermo` to use `externalWallHeatFluxTemperature`.

This solver was initially implemented and tested with OpenFOAM v2212 and later migrated to OpenFOAM 13. The original implementation is made available under [legacy](#) directory and is no longer maintained.

II

Libraries

5	Lagrangian	17
5.1	Parcels	17
6	Thermophysical Models	19
6.1	Thermophysical Properties	19

5. Lagrangian

5.1 Parcels

5.1.1 Extended Thermo Parcel

Provides a variant of `thermoCloud` with varying specific heat.

Important: this extension currently support a single solid phase which needs to be a `tabulatedSolid` as provided in the thermophysical properties below. This will need to be generalized at some point.

Extending a parcel requires several templates to be adapted. The following building blocks are present in the implementation of `extendedThermoCloud`:

- Parcel template class is found in [ExtendedThermoParcel](#)
- The derived parcel instantiation [extendedThermoParcel](#)
- The derived cloud instantiation [extendedThermoCloud](#)

Or in a tree view:

```
parcel
├── clouds
│   └── derived
│       └── extendedThermoCloud
│           ├── extendedThermoCloud.C
│           └── extendedThermoCloud.H
└── parcels
    ├── Templates
    │   ├── ExtendedThermoParcel
    │   │   ├── ExtendedThermoParcel.C
    │   │   ├── ExtendedThermoParcel.H
    │   │   ├── ExtendedThermoParcelI.H
    │   │   ├── ExtendedThermoParcelIO.C
    │   │   ├── ExtendedThermoParcelName.C
    │   └── ExtendedThermoParcelTrackingDataI.H
    └── derived
        └── extendedThermoParcel
            ├── extendedThermoParcel.H
            └── makeExtendedThermoParcelSubmodels.C
```

For using it, append the following to `controlDict` (it depends on extended thermophysical properties so that non-constant properties are supported):

```
libs
(
    "libextendedThermophysicalProperties.so"
    "libextendedLagrangianParcel.so"
);
```


6. Thermophysical Models

6.1 Thermophysical Properties

6.1.1 Polynomial Solid Properties

Library: thermophysicalProperties/solidProperties/polynomialSolidProperties

Extends OpenFOAM's `solidProperties` base class using `Polynomial<8>` to support temperature-dependent 8th-order polynomials density ($\rho(T)$), specific heat capacity ($C_p(T)$), and thermal conductivity (κ).

It is structured as follows:

```
├─ Make
│   └─ files
│       └─ options
├─ polynomialSolid.C
├─ polynomialSolid.H
├─ polynomialSolidI.H
└─ test
    ├── Allclean
    ├── Allrun
    ├── Make
    │   ├── files
    │   └─ options
    └─ testPolynomialSolid.C
```

In `system/controlDict` one must load the compiled library at startup:

```
libs
(
    "libpolynomialSolidProperties.so"
);
```

In `constant/physicalProperties` you can now specify `polynomialSolid` under `solids {}`:

```
solids
{
    polynomialSolid
    {
        molWeight    60.084;
        Hf            0.0;
        emissivity    0.9;

        // Polynomial coefficients for rho(T) = r0 + r1*T + r2*T^2 + ...
        rhoCoeffs     (2500 -0.1 0 0 0 0 0 0);

        // Polynomial coefficients for Cp(T) = c0 + c1*T + c2*T^2 + ...
        CpCoeffs       (800 0.5 -1e-4 0 0 0 0 0);

        // Polynomial coefficients for kappa(T) = c0 + c1*T + c2*T^2 + ...
        kappaCoeffs    (1.5 0 0 0 0 0 0 0);
    }
}
```

6.1.2 Tabulated Solid Properties

Extends OpenFOAM's `solidProperties` base class using `Function1<scalar>` to support tabulated temperature-dependent density ($\rho(T)$), specific heat capacity ($C_p(T)$), and thermal conductivity (κ).

It is structured as follows:

```

├─ Make
│   └─ files
│       └─ options
├─ tabulatedSolid.C
├─ tabulatedSolid.H
├─ tabulatedSolidI.H
└─ test
    ├── Allclean
    ├── Allrun
    ├── Make
    │   ├── files
    │   └─ options
    └─ testTabulatedSolid.C

```

In `system/controlDict` one must load the compiled library at startup:

```

libs
(
    "libtabulatedSolidProperties.so"
);

```

In `constant/physicalProperties` you can now specify `tabulatedSolid` under `solids {}`:

```

solids
{
    tabulatedSolid
    {
        molWeight    60.084;
        Hf            0.0;
        emissivity    0.9;

        // Tabulated density rho(T) [kg/m^3]
        rho
        table
        (
            (300 2500)
            (600 2450)
            (1200 2380)
        );

        // Tabulated specific heat Cp(T) [J/(kg K)]
        Cp
        table
        (
            (300 800)
            (500 950)
            (800 1100)
            (1200 1250)
        );

        // Tabulated thermal conductivity [W/(m K)]
        kappa
        table
        (
            (300 1.5)
            (500 1.5)

```

```
        (800 1.5)
        (1200 1.5)
    );
}
}
```


III

Samples

7	JMAK Solid Kinetics	25
8	Silica Particle Melting	27

7. JMAK Solid Kinetics

About

Sample case can be found [here](#). It illustrates the use of `jmakLaplacianFoam` solver provided in this repository.

Simulate the temperature evolution of a reacting cylinder (diameter 3 inches and height of 6 inches) immersed in coded external convective boundary conditions representing steady air at room temperature. The solid is undergoing volumetric reactions represented by a JMAK kinetics with heat release, leading to a thermal build-up near the sample core. This could represent, *e.g.*, a cement hydration process.

Figure 7.1 illustrates the temperature evolution at a probe placed near the core of the cylinder. The humps are associated to the different reactions that take place over time.

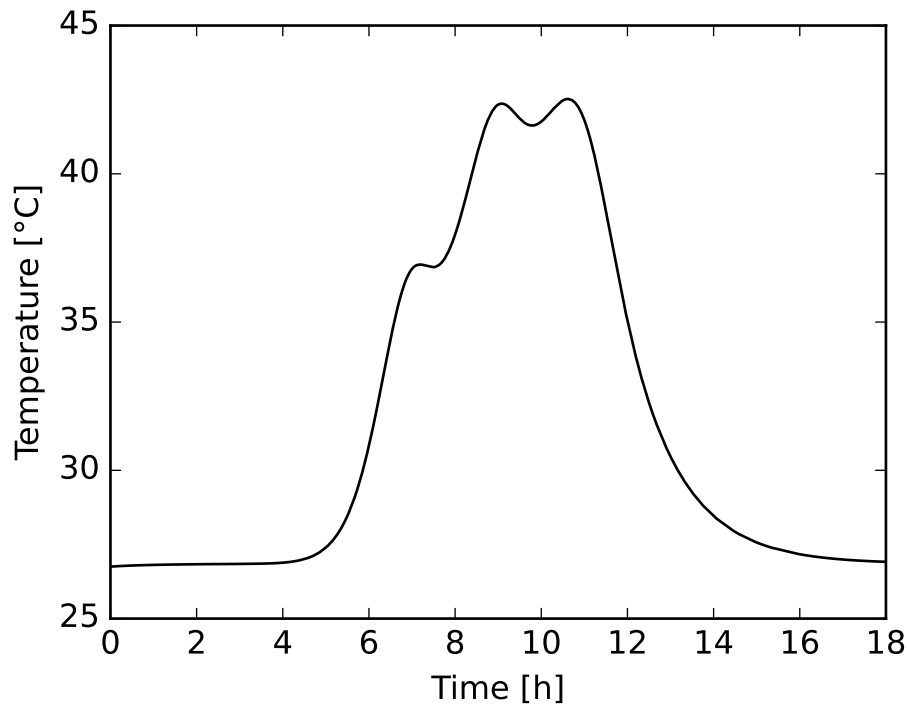


Figure 7.1: Thermal history at probe location.

Figure 7.2 demonstrates how the temperature is distributed in a vertical cut of the cylinder after 8h. Plots of *beta* indicate the JMAK progress of the reaction: we see that reaction 1 has saturated at 1.23, enabling reaction 2 that has already progressed considerably, and reaction 3 is still in earlier stage.

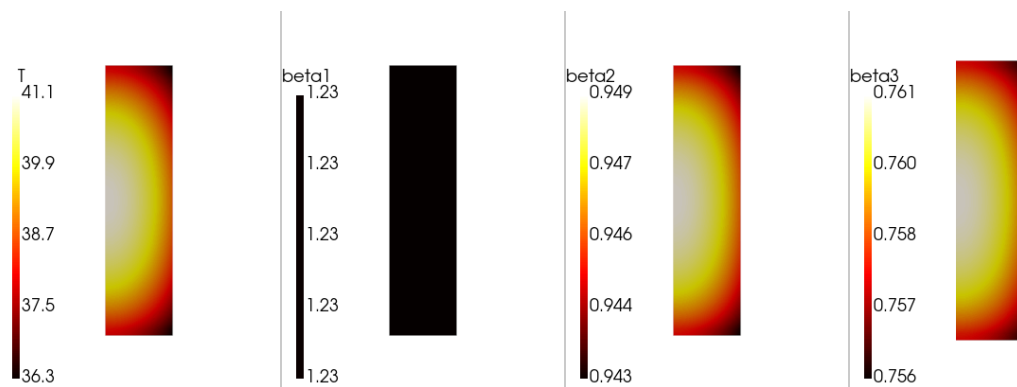


Figure 7.2: Temperature profile at a slice of the domain after 8h.

8. Silica Particle Melting

i About

Sample case can be found [here](#). It intends to illustrate the proper capture of a solid particle melting, as supported by the following libraries provided in this repository:

- `libextendedThermophysicalProperties`: provides `tabulatedSolid`, used for representation of piecewise-linear silica properties, allowing the incorporation of the *lumped* reaction enthalpies in specific heat.
- `libextendedLagrangianParcel`: provides `extendedThermoCloud` allowing the simulation of non-constant properties lagrangian particles.

Simulates the heating of a single silica particle immersed in the flue gases from propane-oxygen combustion. Fluid properties were processed with Cantera and fit to polynomials to match the temperature dependence of adiabatic combustion flue gases (equilibrium computed under constant enthalpy and pressure). Solid specific heat capacity was modified to include smeared latent heat of phase transitions where applicable.

Figure 8.1 displays the thermal history of a single particle under the above conditions. Phase changes produce flat regions, notably the large plateau observed in between 23 – 32 ms associated to the melting of the particle, thus representing the intended goal of capturing the expected silica latent heat of fusion. Since the domain has been intentionally oversized with respect to the required energy to melt the particle, we observe a temperature increase beyond melting point, as expected.

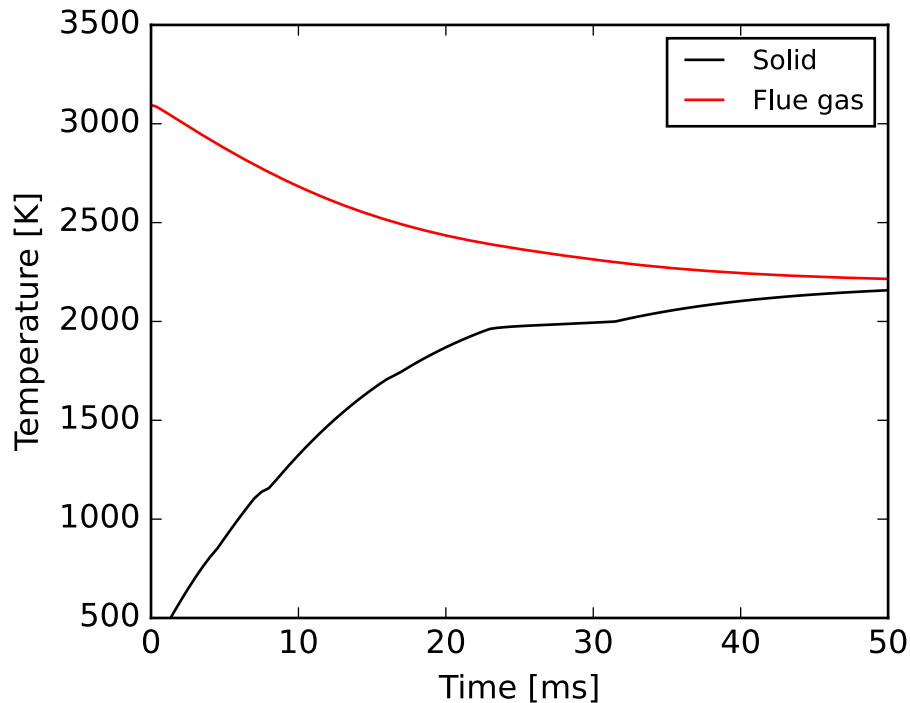


Figure 8.1: Verification of single particle heating thermal history.